Fake News Detection

An End-to-End NLP Pipeline

Headline Classification - IronHack Week 4 NLP Challenge

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Agenda

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Executive Summary

32,206	8	94.94%	0.9288
Training Headlines	Experiments Run	Validation Accuracy	F1 Score

Problem

Classify short news headlines as fake (0) or real (1) - average length only 8 words, so every token counts.

Approach

Single sklearn Pipeline: clean_text() + TF-IDF (1-2 grams) + Linear SVM, tuned with 5-fold GridSearchCV.

Final Model

Linear SVM on TF-IDF (1-2 grams), C=1, max_features=None - small, fast, and the strongest validation score we got.

Outcome

92.63% accuracy on the held-out validation split (6,442 headlines). 9,984 predictions exported to submission.csv.

What we discarded - and why

Deep learning / Neural nets
Classical models reach 94%+. A neural net adds cost with little headroom.

Dense embeddings (Word2Vec, BERT)
Headlines average 8 words; TF-IDF bigrams already capture the discriminative phrases.

Stemming
Chose lemmatisation - keeps tokens human-readable so the SVM coefficients tell a story.

Dataset - Fake News Headlines

34,152 Raw rows in CSV

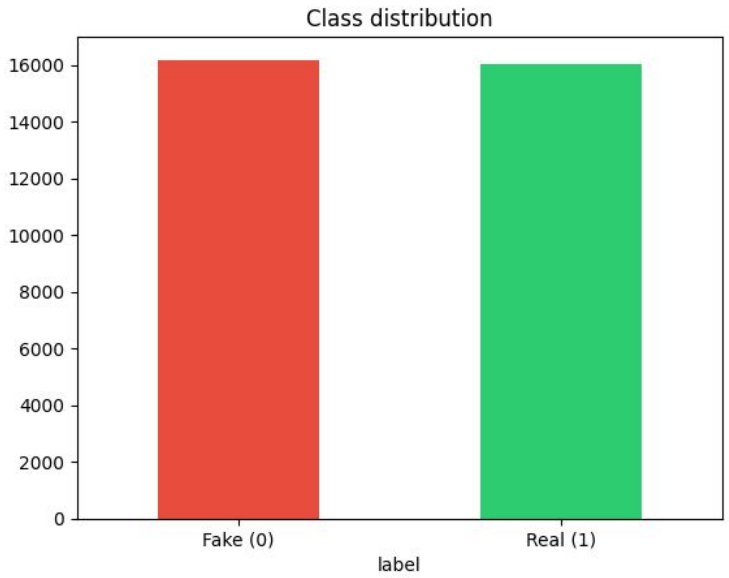
1,946 Duplicates dropped

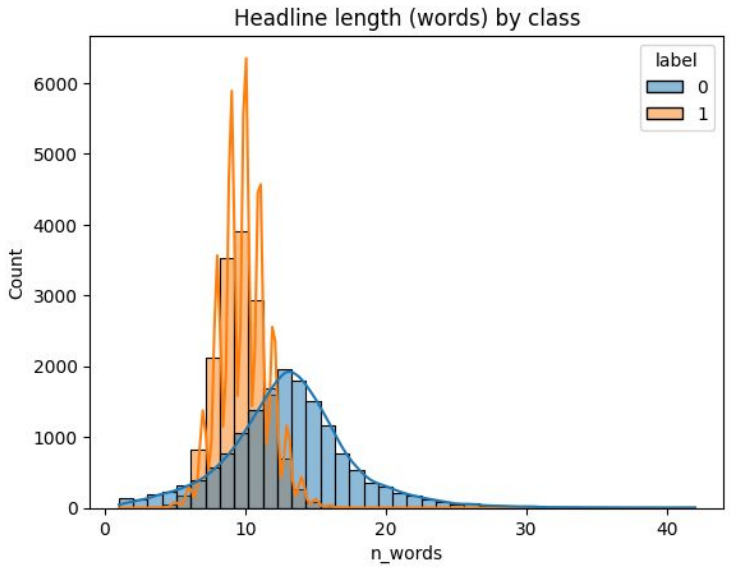
32,206 Final training set

9,984 Unlabelled test set

8 words Average headline

Class balance & headline length





What we learned from EDA

Classes are nearly balanced (~52% real / 48% fake), so plain accuracy is a fair metric. Headlines are short and right-skewed - most under 12 words.

Preprocessing - the clean_text() pipeline

01

Lowercase & strip

Lowercase and drop punctuation/digits. 'Trump!' and 'trump' collapse to one token.

02

Tokenise (NLTK)

word_tokenize splits headlines into tokens, handling contractions .split() misses.

03

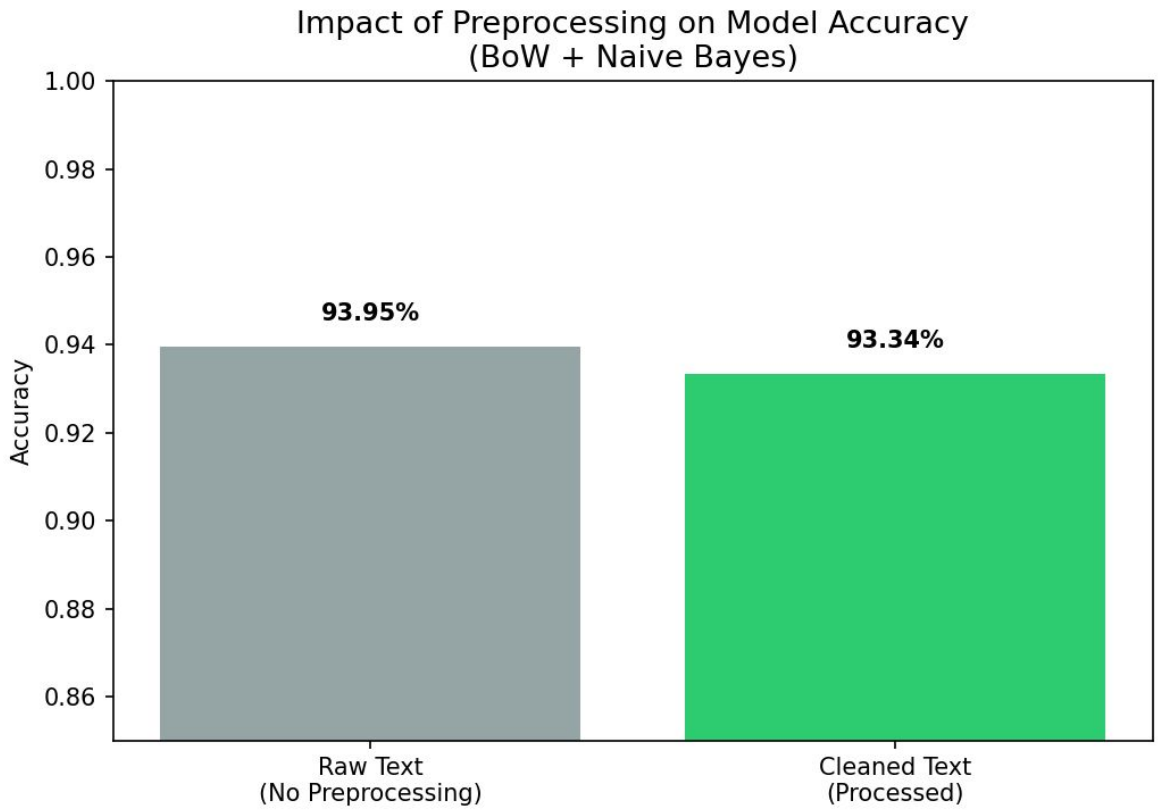
Remove stopwords

Drop English stopwords and tokens under 3 chars to cut noise.

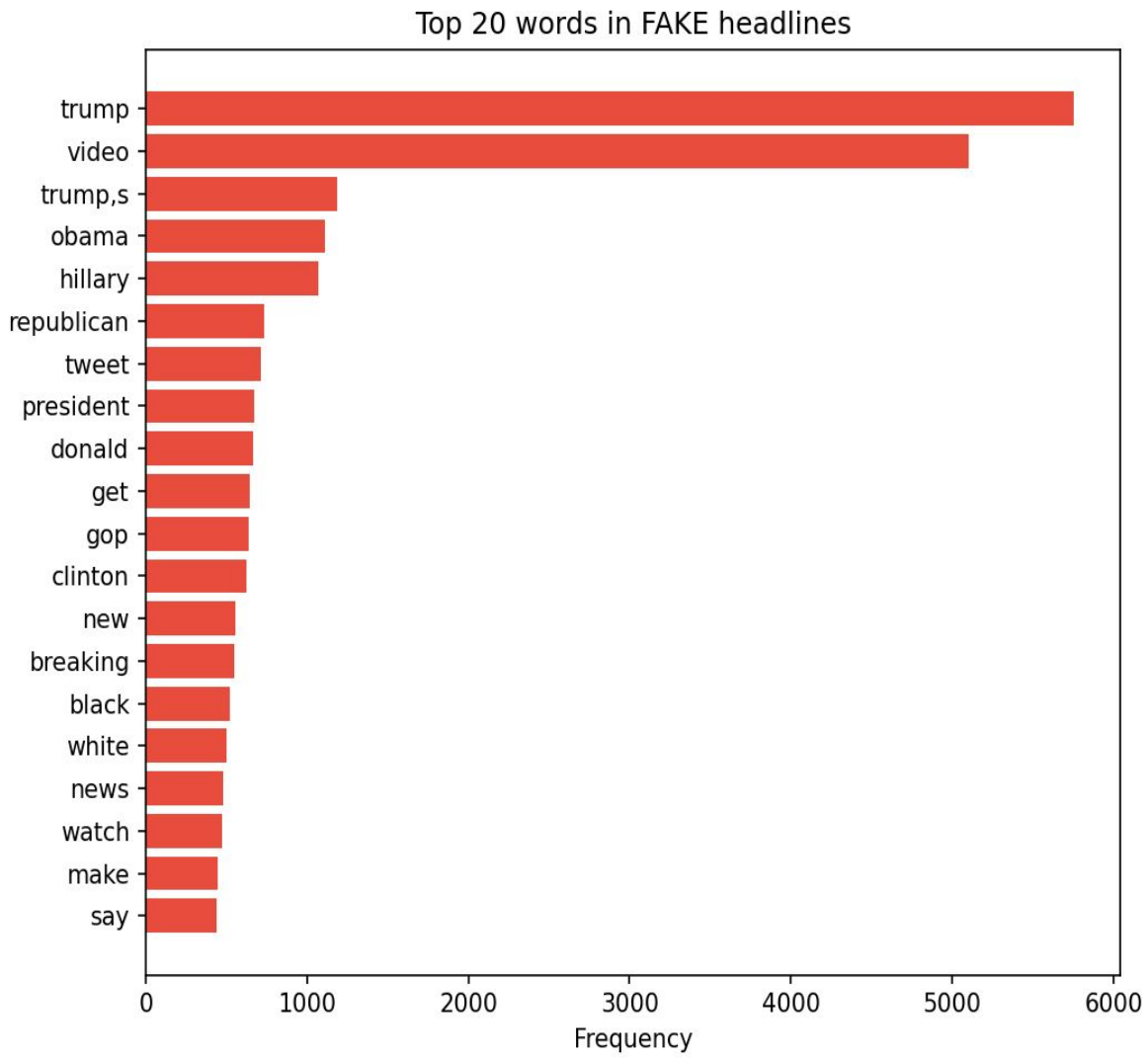
04

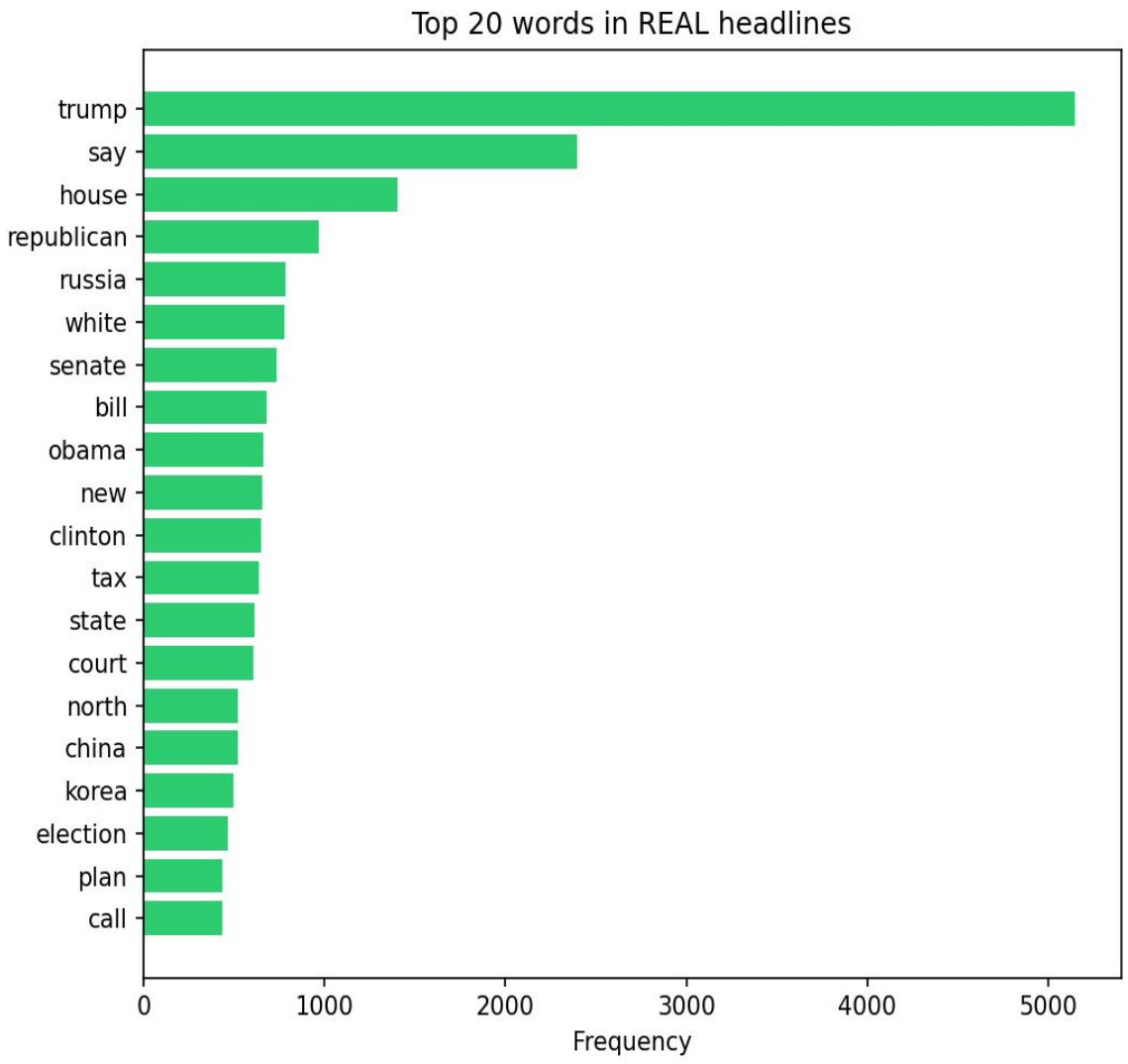
Lemmatise (WordNet)

Lemmatise (running → run). Keeps tokens human-readable for SVM coefficients.



EDA - Most frequent words per class





Feature Extraction - Bag of Words vs TF-IDF

Bag of Words (CountVectorizer)

- Counts raw word occurrences per headline
- No weighting - every token counts equally
- Very fast and memory-efficient
- Used as baseline with Multinomial Naive Bayes

Best validation score

93.95% (RAW + BoW + Naive Bayes)

TF-IDF (TfidfVectorizer) - CHOSEN

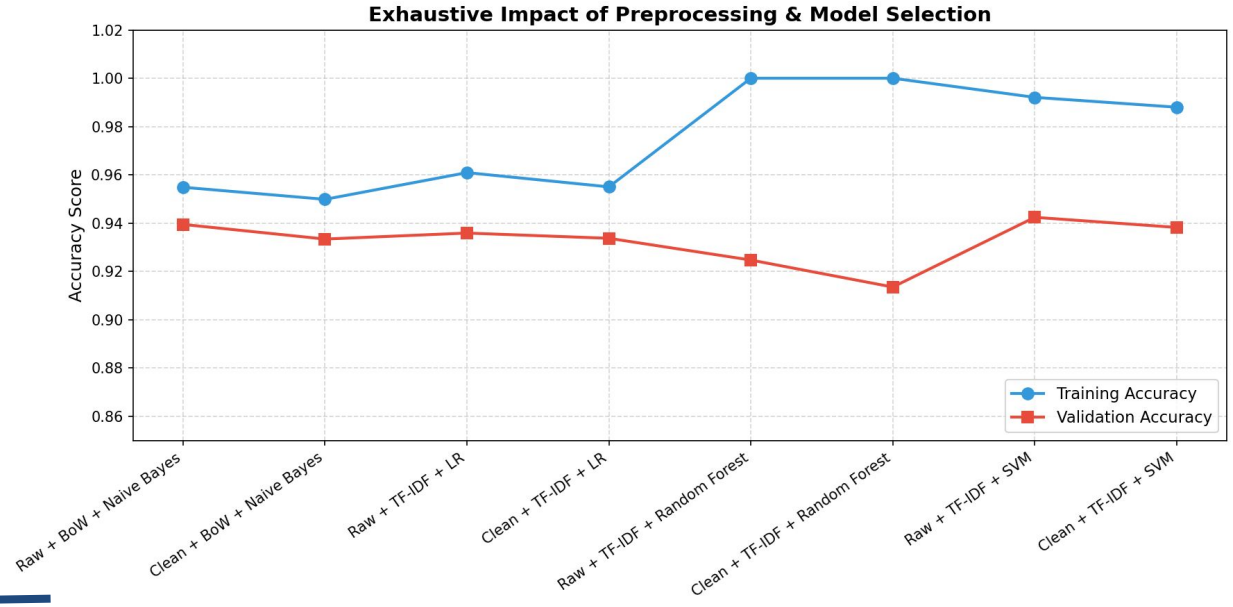
- Term Frequency x Inverse Document Frequency
- Rare/distinctive words get higher weight
- Common words ('says', 'trump') downweighted
- Bigrams (1,2) capture phrases like 'fake news'

Best validation score

94.94% (RAW + TF-IDF + Linear SVM, tuned)

Models - eight experiments compared

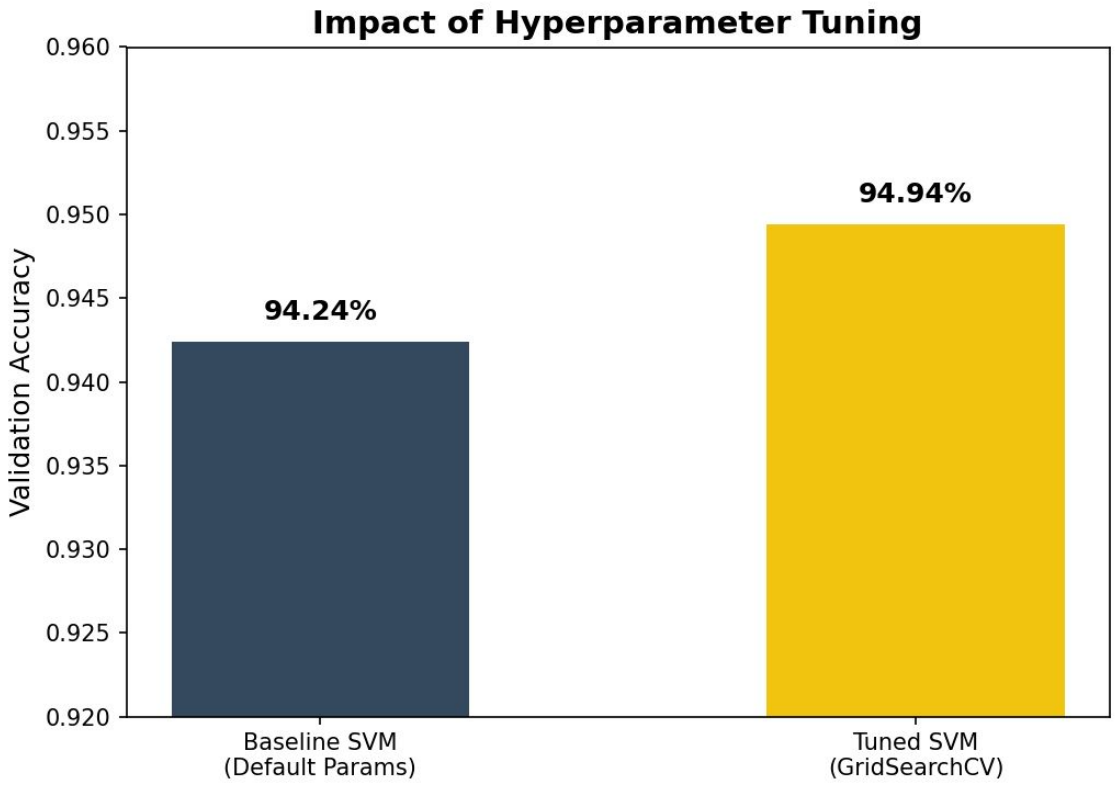
Pipeline	Train	Val
RAW + BoW + Naive Bayes	95.49%	93.95%
Clean + BoW + Naive Bayes	94.99%	93.34%
RAW + TF-IDF + Logistic Reg.	96.09%	93.59%
Clean + TF-IDF + Logistic Reg.	95.50%	93.37%
RAW + TF-IDF + Random Forest	100.0%	92.47%
Clean + TF-IDF + Random Forest	100.0%	91.35%
RAW + TF-IDF + SVM	99.21%	94.24%
Clean + TF-IDF + SVM	98.80%	93.82%



Hyperparameter Tuning - GridSearchCV on TF-IDF + SVM + RAW

94.24%	94.94%	+0.70	5-fold
Before tuning	After tuning	Points gained	Cross-validation

Validation accuracy: baseline vs tuned



Search grid & best parameters

		Best
ngram_range	(1,1) / (1,2) / (1,3)	(1,2)
max_features	5000 / 20000 / None	None
C (LinearSVC)	0.1 / 1 / 10	1

Best Model - what it learned

Final sklearn Pipeline

Input
raw headline string

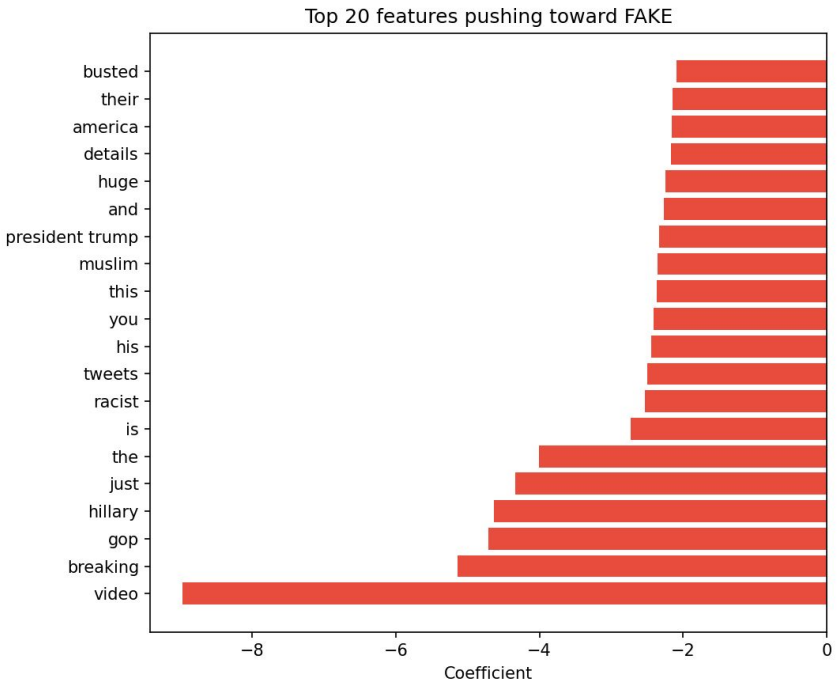
clean_text()
lower, tokenise, stopwords, lemma

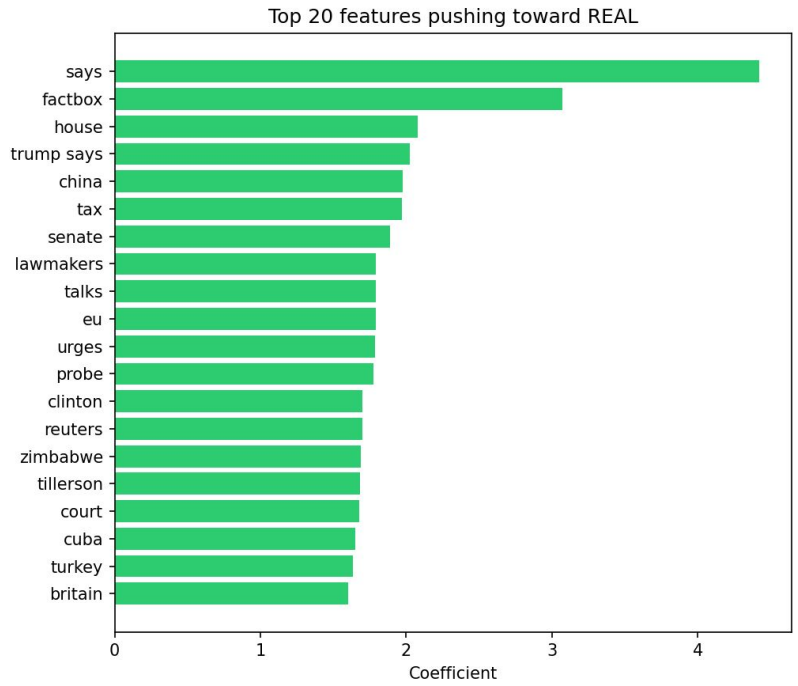
TfidfVectorizer
ngram_range=(1,2), max_features=None

LinearSVC
C=1, hinge loss, L2 penalty

Output
0 = fake | 1 = real

Most informative tokens





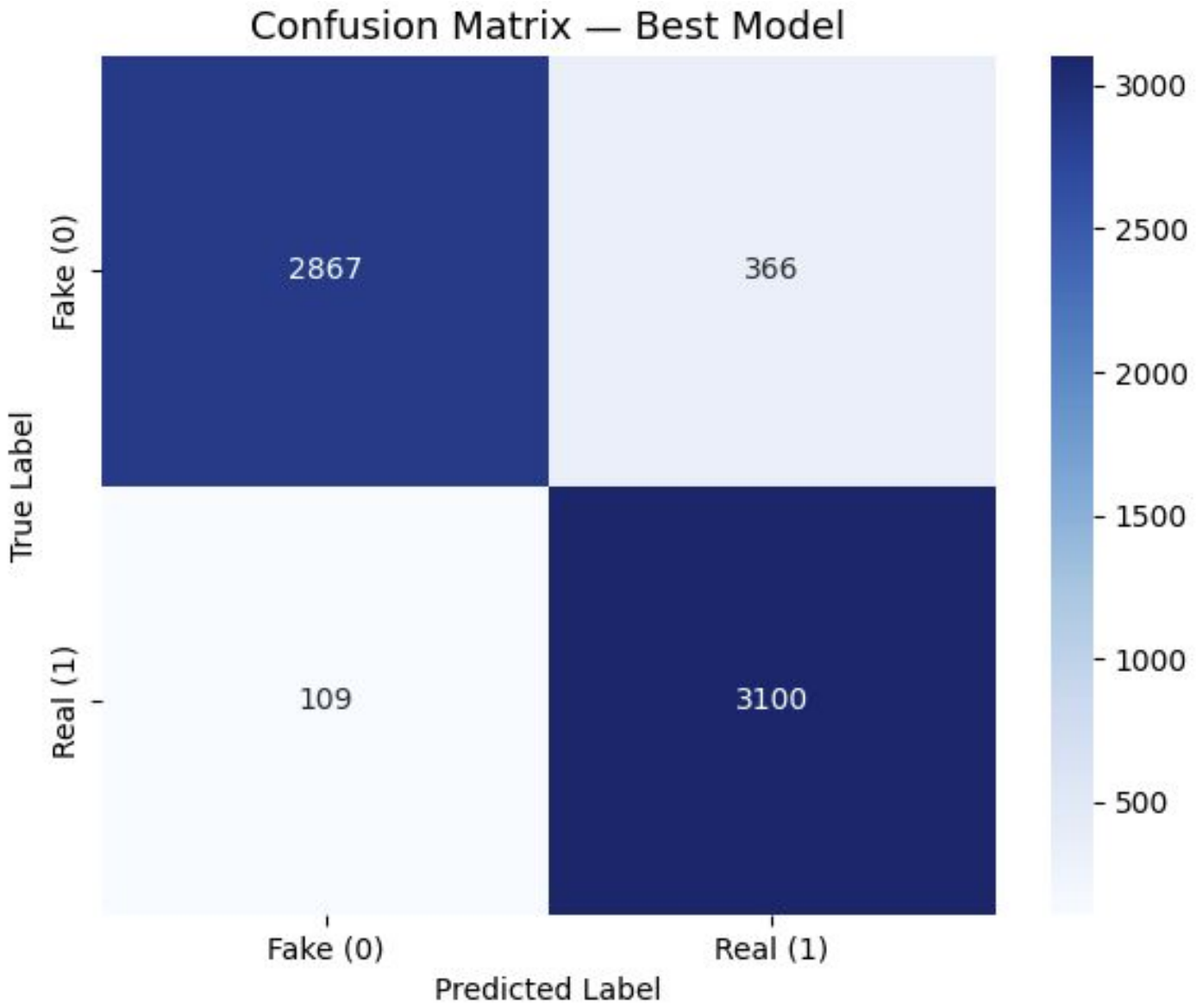
Evaluation - how the winner behaves

94.94%
Accuracy

0.8944
Precision

0.9660
Recall

0.9288
F1 Score



Recap & Key Learnings

CHALLENGES

- Fake news is written to look real
- RF overfit: 100% train vs 91.35% test

RECAP

- 94.94% accuracy
- No embeddings needed
- Unified sklearn Pipeline
- No leakage in Cross Validation

KEY LEARNING

- Tuning: +0.7% gain
- Sometimes simplicity wins

NEXT

- Transformer testing
- Web UI deployment